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Resource Reservation API User Guide

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N/A

Introduction

The following document is the user guide of the REST API for Any management. It includes the model definition as well as all available operations.

SAMPLE USE CASES

Reader will find example of use cases using Usage API in “Open Digital Business Scenarios and Use Cases” document.

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a `BillingAccount` and `SettlementAccount` inheriting properties from the abstract `Account` entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v3.0 Part 2 document.

The `@type` attribute provides a way to represent the actual class type of an entity. For example, within a list of `Account` instances some may be instances of `BillingAccount` where other could be instances of `SettlementAccount`. The `@type` gives this information. All resources and sub-resources of this API have a `@type` attributes that can be provided when this is useful.

The `@referredType` can be used within reference entities (like for instance an `AccountRef` object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the `@type`, when used, denotes the class type of the reference itself, such as `BillingAccountRef` or `SettlementAccountRef`, and not the class type of the referred object. However since reference classes are rarely sub-classed, `@type` is generally not useful in reference objects.

The `@schemaLocation` property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected *characteristics* of an entity.

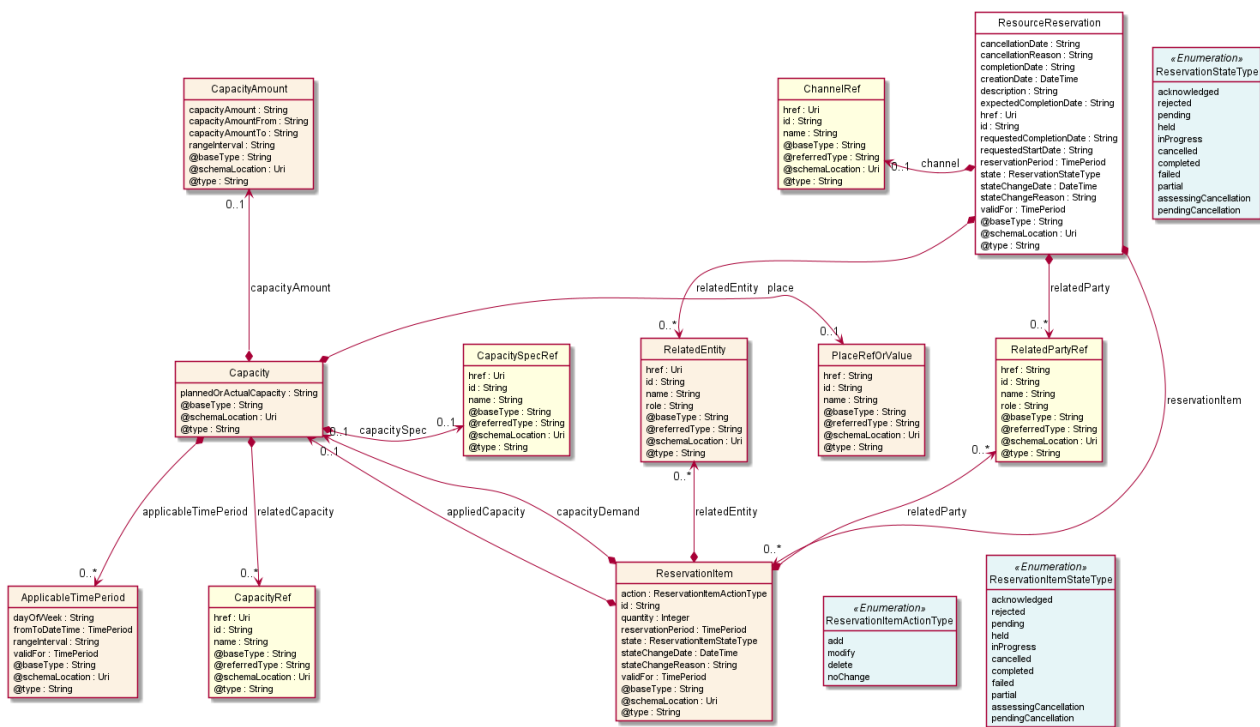
The `@baseType` attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

RESOURCE MODEL

Managed Entity and Task Resource Models

Resource Reservation resource

Resource model



Field descriptions

ResourceReservation fields

cancellationDate	A string. Date when the reservation is cancelled.
cancellationReason	A string. Reason why the reservation is cancelled.
completionDate	A string. Date when the reservation was completed.
creationDate	A date time (DateTime). Date when the reservation was created.
description	A string. free-text description of the reservation.
expectedCompletionDate	A string. Expected completion date amended by the provider.
requestedCompletionDate	A string. Requested completion date from the requestor perspective.

requestedStartDate	A string. Reservation fulfillment start date wished by the requestor. This is used when, for any reason, requestor cannot allow the reservation to begin before a specific date.
stateChangeDate	A date time (DateTime). The date and time the state changed.
stateChangeReason	A string. The reason for changing the state.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
reservationItem	A list of reservation items (ReservationItem [*]).
channel	A channel reference (ChannelRef). The channel to which the resource reference to. e.g. channel for selling product offerings, channel for opening a trouble ticket etc..
relatedEntity	A list of related entities (RelatedEntity [*]). A reference to an entity, where the type of the entity is not known in advance.
relatedParty	A list of related party references (RelatedPartyRef [*]). A related party defines party or party role linked to a specific entity.
reservationItem	A list of reservation items (ReservationItem [*]).
reservationPeriod	A time period. Period of time for which the target entity is to be in a reserved state.
state	A reservation state type (ReservationStateType).
validFor	A time period. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

ApplicableTimePeriod sub-resource

The period of time for which Capacity or CapacityDemand applies.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
dayOfWeek	A string. A day or days representing when the schedule is applicable. For example 2, 3

	represent Monday and Tuesday.
fromToDateTime	A time period. The period of time for which the schedule is applicable.
	Instance values are mutually exclusive with daysOfWeek values.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the from and to DateTime attributes.
	Possible values are "open", "closed", "closedBottom" and "closedTop".
validFor	A time period. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

Capacity sub-resource

Specific ability of an entity measured in quantity and units of quantity over an extended period.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
applicableTimePeriod	A list of applicable time periods (ApplicableTimePeriod [*]). The period of time for which Capacity or CapacityDemand applies.
capacityAmount	A capacity amount (CapacityAmount). quantity that defines the Capacity.
capacitySpec	A capacity spec reference (CapacitySpecRef). capacity specification.
place	A place ref or value (PlaceRefOrValue). Used to define a place useful for the resource.
	Used to indicate reserved resources.
plannedOrActualCapacity	A string. An indicator that specifies whether the capacity is planned or actual.
relatedCapacity	A list of capacity references (CapacityRef [*]). related capacity.

CapacityAmount sub-resource

Quantity that defines the Capacity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

@type	A string. When sub-classing, this defines the sub-class Extensible name.
capacityAmount	A string. A value and units that define the CapacityAmount, such as 10000 ea, 10B Mb. Instance values are mutually exclusive with From and To capacityAmounts and range interval.
capacityAmountFrom	A string. The low range value that a Capacity Amount can take on.
capacityAmountTo	A string. The upper range value that CapacityAmount can take on.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the capacityAmount From and capacityAmountTo attributes. Possible values are "open", "closed", "closedBottom" and "closedTop".

PlaceRefOrValue sub-resource

A place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

RelatedEntity sub-resource

A reference to an entity, where the type of the entity is not known in advance.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. The role of an entity.

ReservationItem sub-resource

An identified part of the reservation. A reservation is decomposed into one or more reservation items.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
action	A reservation item action type (ReservationItemActionType). action to be performed on the entity.
appliedCapacity	A capacity (Capacity). Specific ability of an entity measured in quantity and units of quantity over an extended period.
capacityDemand	A capacity (Capacity). Specific ability of an entity measured in quantity and units of quantity over an extended period.
id	A string. Identifier of the item.
quantity	An integer.
relatedEntity	A list of related entities (RelatedEntity [*]). A reference to an entity, where the type of the entity is not known in advance.
relatedParty	A list of related party references (RelatedPartyRef [*]). A related party defines party or party role linked to a specific entity.
reservationPeriod	A time period. Period of time for which the target entity is to be in a reserved state.
state	A reservation item state type (ReservationItemStateType).
stateChangeDate	A date time (DateTime). The date and time the state changed.
stateChangeReason	A string. The reason for changing the state.
validFor	A time period. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

CapacityRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

CapacitySpecRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ChannelRef relationship

The channel to which the resource reference to. e.g. channel for selling product offerings, channel for opening a trouble ticket etc..

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the channel.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

@schemaLocation An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

@type A string. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRef relationship

RelatedParty reference. A related party defines party or party role linked to a specific entity.

href A string. Reference of the related party, could be a party reference or a party role reference.

id A string. Unique identifier of a related party.

@baseType A string. When sub-classing, this defines the super-class.

@schemaLocation An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

@type A string. When sub-classing, this defines the sub-class Extensible name.

@referredType A string. The actual type of the target instance when needed for disambiguation.

name A string. Name of the related party.

role A string. Role of the related party.

Json representation sample

We provide below the json representation of an example of a 'ResourceReservation' resource object

```
{
  "cancellationDate": "a string ...",
  "cancellationReason": "a string ...",
  "completionDate": "a string ...",
  "creationDate": "2023-06-02T00:00",
  "description": "This resource reservation ...",
  "expectedCompletionDate": "a string ...",
  "requestedCompletionDate": "a string ...",
  "requestedStartDate": "a string ...",
  "stateChangeDate": "2023-06-02T00:00",
  "stateChangeReason": "a string ...",
  "href": "a Uri ...",
  "id": "a Uri ...",
  "reservationItem": [
    {}
  ],
  "channel": {},
  "relatedEntity": [
    {}
  ],
  "relatedParty": [
    {}
  ]
}
```

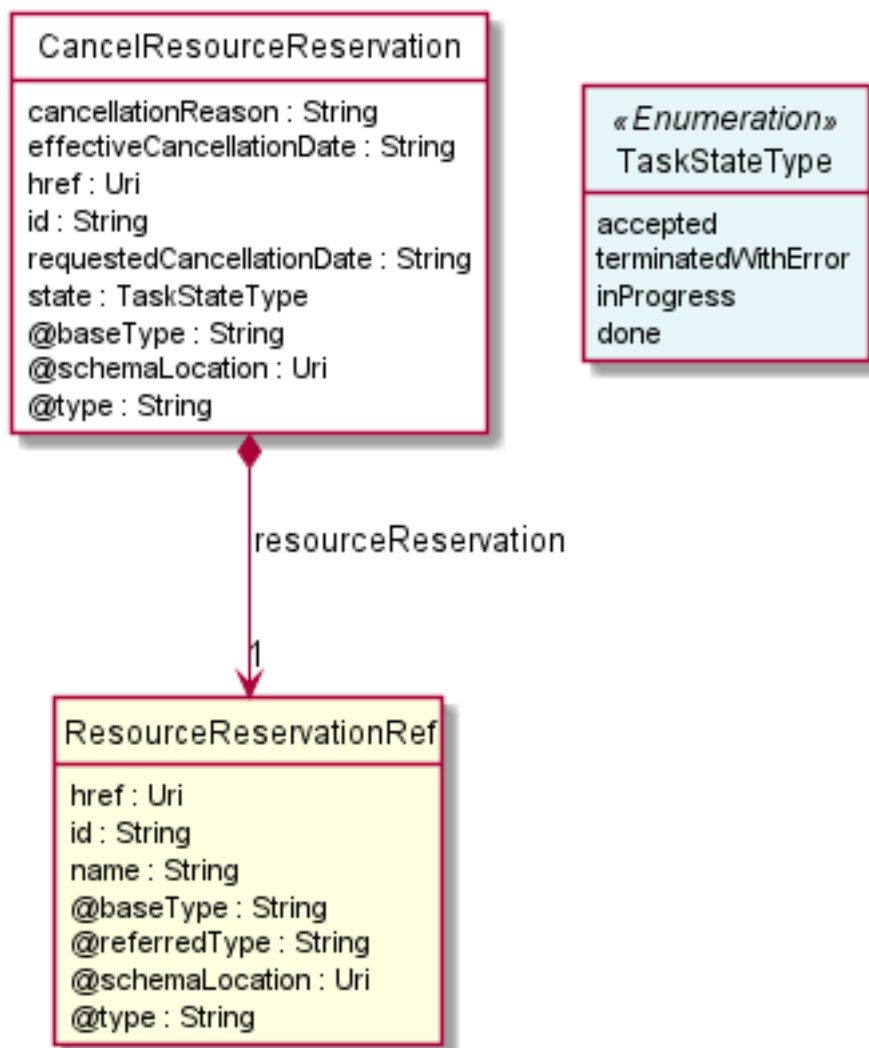
```

},
"reservationPeriod": {},
"state": {},
"validFor": {}
}

```

Cancel Resource Reservation resource

Resource model



Field descriptions

CancelResourceReservation fields

cancellationReason A string. Reason why the reservation is cancelled.

effectiveCancellationDate A string. Date when the reservation is cancelled.

requestedCancellationDate	A string. Date when the submitter wants the reservation to be cancelled.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
resourceReservation	A resource reservation reference (ResourceReservationRef).
state	A task state type (TaskStateType). Possible values for the state of a task.

ResourceReservationRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

We provide below the json representation of an example of a 'CancelResourceReservation' resource object

```
{
  "cancellationReason": "a string ...",
  "effectiveCancellationDate": "a string ...",
  "requestedCancellationDate": "a string ...",
  "href": "a Uri ...",
  "id": "a Uri ...",
  "resourceReservation": {},
  "state": {}
}
```

Notification Resource Models

8 notifications are defined for this API

Notifications related to ResourceReservation:

- ResourceReservationCreateEvent
- ResourceReservationAttributeValueChangeEvent
- ResourceReservationDeleteEvent
- ResourceReservationStateChangeEvent
- ResourceReservationInformationRequiredEvent

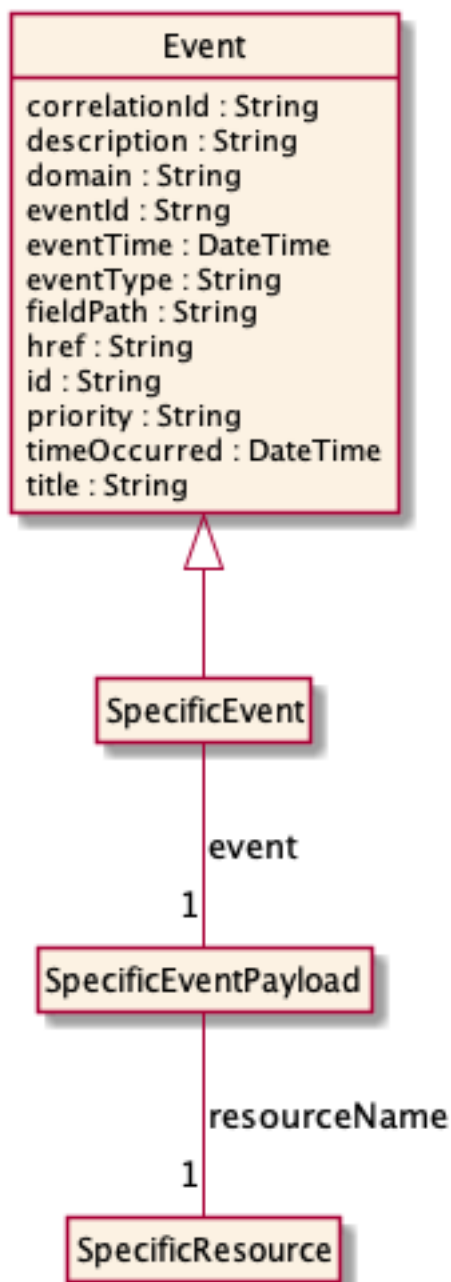
Notifications related to CancelResourceReservation:

- CancelResourceReservationCreateEvent
- CancelResourceReservationStateChangeEvent
- CancelResourceReservationInformationRequiredEvent

The notification structure for all notifications in this API follow the pattern depicted by the figure below.

A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType).

This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).



Resource Reservation Create Event

Notification `ResourceReservationCreateEvent` case for resource `ResourceReservation`

Json representation sample

We provide below the json representation of an example of a 'ResourceReservationCreateEvent' notification event object

```
{
  "eventId": "00001",
  "eventTime": "2015-11-16T16:42:25-04:00",
```

```
"eventType": "ResourceReservationCreateEvent",
"event": {
  "resourceReservation" :
    {-- SEE ResourceReservation RESOURCE SAMPLE --}
}
}
```

Resource Reservation Attribute Value Change Event

Notification ResourceReservationAttributeValueChangeEvent case for resource ResourceReservation

Json representation sample

We provide below the json representation of an example of a 'ResourceReservationAttributeValueChangeEvent' notification event object

```
{
  "eventId": "00001",
  "eventTime": "2015-11-16T16:42:25-04:00",
  "eventType": "ResourceReservationAttributeValueChangeEvent",
  "event": {
    "resourceReservation" :
      {-- SEE ResourceReservation RESOURCE SAMPLE --}
  }
}
```

Resource Reservation Delete Event

Notification ResourceReservationDeleteEvent case for resource ResourceReservation

Json representation sample

We provide below the json representation of an example of a 'ResourceReservationDeleteEvent' notification event object

```
{
  "eventId": "00001",
  "eventTime": "2015-11-16T16:42:25-04:00",
  "eventType": "ResourceReservationDeleteEvent",
  "event": {
    "resourceReservation" :
      {-- SEE ResourceReservation RESOURCE SAMPLE --}
  }
}
```

Resource Reservation State Change Event

Notification ResourceReservationStateChangeEvent case for resource ResourceReservation

Json representation sample

We provide below the json representation of an example of a 'ResourceReservationStateChangeEvent' notification event object

```
{
  "eventId": "00001",
  "eventTime": "2015-11-16T16:42:25-04:00",
  "eventType": "ResourceReservationStateChangeEvent",
  "event": {
    "resourceReservation":
      {-- SEE ResourceReservation RESOURCE SAMPLE --}
  }
}
```

Resource Reservation Information Required Event

Notification ResourceReservationInformationRequiredEvent case for resource ResourceReservation

Json representation sample

We provide below the json representation of an example of a 'ResourceReservationInformationRequiredEvent' notification event object

```
{
  "eventId": "00001",
  "eventTime": "2015-11-16T16:42:25-04:00",
  "eventType": "ResourceReservationInformationRequiredEvent",
  "event": {
    "resourceReservation":
      {-- SEE ResourceReservation RESOURCE SAMPLE --}
  }
}
```

Cancel Resource Reservation Create Event

Notification CancelResourceReservationCreateEvent case for resource CancelResourceReservation

Json representation sample

We provide below the json representation of an example of a 'CancelResourceReservationCreateEvent' notification event object

```
{
  "eventId": "00001",
  "eventTime": "2015-11-16T16:42:25-04:00",
  "eventType": "CancelResourceReservationCreateEvent",
  "event": {
    "cancelResourceReservation":
      {-- SEE CancelResourceReservation RESOURCE SAMPLE --}
  }
}
```

```
}  
}
```

Cancel Resource Reservation State Change Event

Notification CancelResourceReservationStateChangeEvent case for resource CancelResourceReservation

Json representation sample

We provide below the json representation of an example of a 'CancelResourceReservationStateChangeEvent' notification event object

```
{  
  "eventId": "00001",  
  "eventTime": "2015-11-16T16:42:25-04:00",  
  "eventType": "CancelResourceReservationStateChangeEvent",  
  "event": {  
    "cancelResourceReservation":  
      {-- SEE CancelResourceReservation RESOURCE SAMPLE --}  
  }  
}
```

Cancel Resource Reservation Information Required Event

Notification CancelResourceReservationInformationRequiredEvent case for resource CancelResourceReservation

Json representation sample

We provide below the json representation of an example of a 'CancelResourceReservationInformationRequiredEvent' notification event object

```
{  
  "eventId": "00001",  
  "eventTime": "2015-11-16T16:42:25-04:00",  
  "eventType": "CancelResourceReservationInformationRequiredEvent",  
  "event": {  
    "cancelResourceReservation":  
      {-- SEE CancelResourceReservation RESOURCE SAMPLE --}  
  }  
}
```

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Complete Update of an Entity	PUT Resource	PUT must be used to completely update a resource identified by its resource URI
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on Resource Reservation

List resource reservations

GET /resourceReservation?fields=...&{filtering}

Description

This operation list resource reservation entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving ResourceReservation resources.

Request
GET /tmf-api/resourceReservation/v4/resourceReservation Accept: application/json
Response
200 [{ "cancellationDate": "a string ...", "cancellationReason": "a string ...", "completionDate": "a string ...", "creationDate": "2023-06-02T00:00", "description": "This resource reservation ...", "expectedCompletionDate": "a string ...", "requestedCompletionDate": "a string ...", "requestedStartDate": "a string ...", "stateChangeDate": "2023-06-02T00:00", "stateChangeReason": "a string ...", "href": "a Uri ...", "id": "a Uri ...", "reservationItem": [{}], "channel": {}, "relatedEntity": [{}], "relatedParty": [{}], },]

```

"reservationPeriod": {},
"state": {},
"validFor": {}
}
]

```

Retrieve resource reservation

GET /resourceReservation/{id}?fields=...&{filtering}

Description

This operation retrieves a resource reservation entity.

Attribute selection is enabled for all first level attributes.

Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a ResourceReservation resource.

Request
GET /tmf-api/resourceReservation/v4/resourceReservation/a Uri ... Accept: application/json
Response
200 { "cancellationDate": "a string ...", "cancellationReason": "a string ...", "completionDate": "a string ...", "creationDate": "2023-06-02T00:00", "description": "This resource reservation ...", "expectedCompletionDate": "a string ...", "requestedCompletionDate": "a string ...", "requestedStartDate": "a string ...", "stateChangeDate": "2023-06-02T00:00", "stateChangeReason": "a string ...", "href": "a Uri ...", "id": "a Uri ...", "reservationItem": [{}], "channel": {},

```

"relatedEntity": [
  {}
],
"relatedParty": [
  {}
],
"reservationPeriod": {},
"state": {},
"validFor": {}
}

```

Create resource reservation

POST /resourceReservation

Description

This operation creates a resource reservation entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ResourceReservation, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
----------------------	------

Non Mandatory Attributes	Rule
cancellationDate	
cancellationReason	
completionDate	
creationDate	
description	
expectedCompletionDate	
requestedCompletionDate	
requestedStartDate	
stateChangeDate	
stateChangeReason	
@baseType	
@schemaLocation	
@type	
reservationItem	
channel	
relatedEntity	
relatedParty	
reservationItem	
reservationPeriod	
state	

validFor	
----------	--

Usage Samples

Here's an example of a request for creating a ResourceReservation resource. In this example the request only passes mandatory attributes.

Request
POST /tmf-api/resourceReservation/v4/resourceReservation Content-Type: application/json {}
Response
201 { "href": "a Uri ...", "id": "a Uri ..." }

Patch resource reservation

PATCH /resourceReservation/{id}

Description

This operation allows partial updates of a resource reservation entity. Support of json/merge (<https://tools.ietf.org/html/rfc7386>) is mandatory, support of json/patch (<http://tools.ietf.org/html/rfc5789>) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
cancellationDate	
cancellationReason	

completionDate	
creationDate	
description	
expectedCompletionDate	
requestedCompletionDate	
requestedStartDate	
stateChangeDate	
stateChangeReason	
@baseType	
@schemaLocation	
@type	
reservationItem	
channel	
relatedEntity	
relatedParty	
reservationItem	
reservationPeriod	
state	
validFor	

Non Patchable Attributes	Rule
id	
href	

Usage Samples

Here's an example of a request for patching a ResourceReservation resource.

Request
<pre>PATCH /tmf-api/resourceReservation/v4/resourceReservation/a Uri ... Content-Type: application/merge-patch+json { "name": "new name" }</pre>
Response
<pre>200 { "cancellationDate": "a string ...", "cancellationReason": "a string ...", "completionDate": "a string ...",</pre>

```

"creationDate": "2023-06-02T00:00",
"description": "This resource reservation ...",
"expectedCompletionDate": "a string ...",
"requestedCompletionDate": "a string ...",
"requestedStartDate": "a string ...",
"stateChangeDate": "2023-06-02T00:00",
"stateChangeReason": "a string ...",
"href": "a Uri ...",
"id": "a Uri ...",
"reservationItem": [
  {}
],
"channel": {},
"relatedEntity": [
  {}
],
"relatedParty": [
  {}
],
"reservationPeriod": {},
"state": {},
"validFor": {},
"name": "new name"
}

```

Delete resource reservation

DELETE /resourceReservation/{id}

Description

This operation deletes a resource reservation entity.

Usage Samples

Here's an example of a request for deleting a ResourceReservation resource.

Request
DELETE /tmf-api/resourceReservation/v4/resourceReservation/42
Response
204

Operations on Cancel Resource Reservation

List cancel resource reservations

GET /cancelResourceReservation?fields=...&{filtering}

Description

This operation list cancel resource reservation entities.

Attribute selection is enabled for all first level attributes.

Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving CancelResourceReservation resources.

Request
GET /tmf-api/resourceReservation/v4/cancelResourceReservation Accept: application/json
Response
200 [{ "cancellationReason": "a string ...", "effectiveCancellationDate": "a string ...", "requestedCancellationDate": "a string ...", "href": "a Uri ...", "id": "a Uri ...", "resourceReservation": {}, "state": {} }]

Retrieve cancel resource reservation

GET /cancelResourceReservation/{id}?fields=...&{filtering}

Description

This operation retrieves a cancel resource reservation entity.

Attribute selection is enabled for all first level attributes.

Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a CancelResourceReservation resource.

Request
GET /tmf-api/resourceReservation/v4/cancelResourceReservation/a Uri ... Accept: application/json
Response
200 <pre>{ "cancellationReason": "a string ...", "effectiveCancellationDate": "a string ...", "requestedCancellationDate": "a string ...", "href": "a Uri ...", "id": "a Uri ...", "resourceReservation": {}, "state": {} }</pre>

Create cancel resource reservation

POST /cancelResourceReservation

Description

This operation creates a cancel resource reservation entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a CancelResourceReservation, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
ResourceReservation	
resourceReservation	

Non Mandatory Attributes	Rule
cancellationReason	
effectiveCancellationDate	
requestedCancellationDate	
@baseType	
@schemaLocation	
@type	
state	

Additional Rules

The following table provides additional rules indicating mandatory fields in sub-resources or relationships when creating a CancelResourceReservation resource.

Context	Mandatory Sub-Attributes
ResourceReservation	id

The following pre-conditions apply for this operation.

Pre-conditions
resource reservation id must exist
When creating a cancel resource reservation (post) the id, href, effectiveCancellationDate and state fields must not be valued in the request

Usage Samples

Here's an example of a request for creating a CancelResourceReservation resource. In this example the request only passes mandatory attributes.

Request
POST /tmf-api/resourceReservation/v4/cancelResourceReservation Content-Type: application/json <pre>{ "resourceReservation": {} }</pre>
Response
201 <pre>{ "href": "a Uri ...",</pre>

```
"id": "a Uri ...",  
"resourceReservation": {}  
}
```

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request
POST /api/hub Accept: application/json {"callback": "http://in.listener.com"}
Response
201 Content-Type: application/json Location: /api/hub/42 {"id": "42", "callback": "http://in.listener.com", "query": null}

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request
DELETE /api/hub/42 Accept: application/json
Response
204

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example “EVENT TYPE” should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request
<pre>POST /client/listener Accept: application/json { "event": { EVENT BODY }, "eventType": "EVENT_TYPE" }</pre>
Response
201

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Release 1.0	04/15/2017	Pierre Gauthier TM Forum pgauthier@tmforum.org Mariano Belaunde Orange Labs	First Release of the Document.
Release 2.0	11/06/2018	Mariano Belaunde Orange Labs	Alignment with Guidelines 3.0

Contributors to Document

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